

Iron Deficiency Anemia (IDA): Comprehensive Clinical Review with Oral Iron Strategies, CKD & Dialysis Guidelines

About the Author

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Dr. Hamid Fazal has extensive clinical and academic experience in internal medicine, nephrology-associated care, nutrition, and critical care medicine. His professional experience includes:

4 years of experience working in Intensive Care Units (ICU)

3 years of experience working in Dialysis Units and Chronic Kidney Disease (CKD) outpatient services

Served as CKD OPD Incharge

Participated in extensive research work related to Aging and Dementia at the NYU Aging and Dementia Research Center, United States

His areas of clinical interest include:

Chronic Kidney Disease (CKD)

Dialysis medicine

Nutritional medicine

Geriatric medicine

Neurodegenerative disorders

Internal medicine education

1. Definition

Iron deficiency anemia = anemia caused by insufficient iron available for erythropoiesis and hemoglobin synthesis.

Typical laboratory profile:

Test

Typical Finding

Hemoglobin

Low

MCV

Low (microcytic)

MCH

Low

Ferritin

Low

Serum iron

Low

TIBC/transferrin

High

Transferrin saturation (TSAT)

Low

RDW

Often elevated

Ferritin:

<15 ng/mL = highly specific

<30 ng/mL = strongly suggestive in most adults

Inflammation complicates interpretation because ferritin is an acute-phase reactant.

In inflammatory states:

Ferritin <100 ng/mL + TSAT <20% may still indicate iron deficiency.

2. Iron Physiology (High Yield)

Total body iron:

~3–4 g in adults

Distribution:

Hemoglobin: ~65%

Ferritin/hemosiderin stores

Myoglobin

Enzymes

Absorption occurs mainly in:

duodenum

proximal jejunum

Daily iron absorption:

normally ~1–2 mg/day

3. Hepcidin Physiology (Modern Core Concept)

Hepcidin is the master iron regulatory hormone produced by the liver.

It:

blocks ferroportin

decreases intestinal absorption

traps iron in macrophages

Hepcidin increases with:

inflammation

CKD

infection

repeated oral iron doses

This explains:

poor absorption in inflammatory states

benefit of alternate-day dosing.

4. Causes of Iron Deficiency

A. Blood Loss (Most Common)

GI causes

Peptic ulcer disease

Gastritis

NSAIDs

GI malignancy

IBD

Hemorrhoids

Angiodysplasia

Gynecologic

Menorrhagia

Fibroids

Other

Blood donation

Repeated phlebotomy

Important rule: Iron deficiency in:

adult men

postmenopausal women

requires GI evaluation until proven otherwise.

B. Increased Demand

Pregnancy

Lactation

Adolescence

Rapid growth

C. Reduced Absorption

Celiac disease

Bariatric surgery

H. pylori

Atrophic gastritis

Chronic PPI therapy

D. Nutritional Deficiency

Malnutrition

Restrictive diets

Elderly frailty

5. Clinical Features

General anemia symptoms

Fatigue

Weakness

Dyspnea

Palpitations

Reduced exercise tolerance

Iron-specific symptoms

Pica

Pagophagia

Restless legs syndrome

Hair loss

Glossitis

Angular cheilitis

Koilonychia

6. Diagnosis

Initial tests

CBC

Ferritin

Iron studies

Reticulocyte count

Typical pattern

Parameter

IDA

Hb

Low

MCV

Low

Ferritin

Low

TSAT

Low

TIBC

High

7. Differential Diagnosis

Condition

Ferritin

TIBC

MCV

Iron deficiency anemia

Low

High

Low

Anemia of chronic disease

Normal/high

Low/normal

Normal/low

Thalassemia trait

Normal

Normal

Very low

8. Oral Iron Therapy — Modern Evidence-Based Approach

Historical approach

Previously:

iron given BID/TID

Modern evidence shows:

lower doses absorb better

fewer side effects

improved adherence

because repeated dosing increases hepcidin.

9. Optimal Oral Iron Strategy

Preferred elemental iron dose

Most adults: 40–80 mg elemental iron:

once daily OR

alternate day

Often superior to:

200–300 mg/day divided dosing.

10. Oral Iron Preparations

A. Ferrous Sulfate (Preferred First-Line)

Example

325 mg tablet $\approx$ 65 mg elemental iron.

Advantages

cheapest

strongest evidence

widely available

Disadvantages

GI intolerance common

B. Ferrous Fumarate

325 mg $\approx$ 106 mg elemental iron.

Useful when:

fewer tablets desired.

C. Ferrous Gluconate

325 mg $\approx$ 35 mg elemental iron.

May be:

better tolerated

less constipation

D. Polysaccharide Iron Complex

Marketed as gentler.

Evidence for superiority:

inconsistent.

More expensive.

E. Ferric Maltol

Useful especially in:

IBD

oral iron intolerance

Potentially less GI irritation.

F. Heme Iron Polypeptide

Theoretical better absorption.

Limitations:

expensive

limited evidence advantage

11. Which Oral Iron Is Best?

Practical recommendation

First-line for most patients

Ferrous sulfate.

Because:

excellent evidence

inexpensive

highly effective

If intolerance develops

Before stopping iron:

Lower dose

Alternate-day regimen

Take with small meal

Switch salt (gluconate/fumarate)

Consider ferric maltol

12. How to Take Iron Correctly

Best absorption

Take:

on empty stomach

with water

optionally with vitamin C

Reduced absorption by

Factor

Examples

Calcium

Milk, supplements

PPIs

Omeprazole

Antacids

Calcium carbonate

Tea/coffee

Tannins

Cereals/phytates

High-fiber foods

Separate by:

2–4 hours.

13. Common Side Effects

Constipation

Nausea

Abdominal discomfort

Metallic taste

Dark stool

14. Managing GI Side Effects

Best strategy:

reduce dose

switch to alternate-day dosing

use smaller elemental iron dose

try different preparation

Many patients tolerate: lower-dose regimens better than high-dose traditional regimens.

15. Expected Hematologic Response

Reticulocytosis

Begins: ~5–10 days

Hemoglobin rise

~1–2 g/dL every 2–3 weeks

Ferritin recovery

Requires: months.

16. Duration of Therapy

Important principle: Do NOT stop when Hb normalizes.

Continue: ~3 months after normalization

to replenish stores.

17. Causes of Failure of Oral Iron

Nonadherence

Ongoing bleeding

Wrong diagnosis

Malabsorption

Inflammation/hepcidin excess

Celiac disease

H. pylori

CKD-associated functional deficiency

18. Indications for IV Iron

Major indications

Severe oral intolerance

Malabsorption

Bariatric surgery

Active IBD

CKD with ESA use

Dialysis

Need for rapid correction

Ongoing blood loss

19. IV Iron Preparations

Formulation

Notes

Iron sucrose

Common, well-established

Ferric carboxymaltose

Large single doses

Ferumoxytol

Rapid administration

Iron dextran (low MW)

Total-dose infusion possible

Modern IV iron:

substantially safer than older formulations.

20. Risks of IV Iron

Potential adverse effects:

infusion reactions

hypotension

rare anaphylaxis

hypophosphatemia (especially ferric carboxymaltose)

21. CKD and Dialysis Patients — Current Guideline-Based Approach

Why iron deficiency is common in CKD

Mechanisms:

chronic inflammation → increased hepcidin

reduced GI absorption

blood loss during dialysis

repeated blood sampling

ESA-induced iron utilization

Many CKD patients have: functional iron deficiency.

Meaning:

iron stores may appear “adequate”

but iron unavailable for erythropoiesis.

22. Definitions in CKD

Absolute iron deficiency

Typically:

TSAT <20%

ferritin <100 ng/mL (non-dialysis/peritoneal dialysis)

ferritin <200 ng/mL (hemodialysis)

Functional iron deficiency

ferritin normal/high

TSAT low (<20%)

due to inflammation/hepcidin excess.

23. KDIGO-Oriented Iron Therapy Thresholds

Current guideline principles (KDIGO-based):

Consider iron therapy if:

TSAT $\leq$ 30% AND

ferritin $\leq$ 500 ng/mL

especially when:

increasing Hb desired

reducing ESA dose desired

avoiding transfusion desired.

24. Oral vs IV Iron in CKD

Non-dialysis CKD

Either:

oral iron OR

IV iron

depending on:

severity

tolerance

urgency

inflammation

Oral iron may work in:

early CKD

mild deficiency

But absorption is often poor.

Hemodialysis patients

IV iron is generally preferred.

Reasons:

poor oral absorption

ongoing blood loss

inflammation

ESA use

Most dialysis patients respond substantially better to IV iron.

25. Common IV Iron Regimens in Dialysis

Frequently used agents:

iron sucrose

ferric gluconate

Common strategy: small repeated maintenance doses during dialysis sessions.

Example:

iron sucrose 50–100 mg intermittently

(protocols vary).

26. ESA (Erythropoiesis-Stimulating Agents)

Examples:

epoetin alfa

darbepoetin alfa

Iron deficiency must be corrected for optimal ESA response.

Without iron:

ESA resistance occurs

escalating ESA doses become ineffective and risky.

27. Hemoglobin Targets in CKD

Avoid normalization to high Hb.

Most guidelines: target Hb approximately: 10–11.5 g/dL

Higher targets linked with:

stroke

hypertension

thrombosis

cardiovascular events

when ESA aggressively used.

28. Monitoring in CKD

Typical monitoring:

Hb

ferritin

TSAT

every:

1–3 months in dialysis patients

less frequently in stable non-dialysis CKD

29. Iron Overload Risk in CKD

Avoid excessive iron administration.

Potential concerns:

oxidative stress

infection risk

iron overload

Ferritin persistently >800–1000 ng/mL: usually prompts reassessment.

30. Pregnancy and Iron Deficiency

First-line: oral iron unless:

severe anemia

intolerance

urgent late gestation correction needed

IV iron increasingly used in: 2nd and 3rd trimester when indicated.

31. Heart Failure

Iron deficiency itself worsens outcomes.

IV ferric carboxymaltose may improve:

exercise tolerance

symptoms

quality of life

especially in HFrEF.

32. Blood Transfusion

Transfusion: is NOT definitive iron therapy.

Reserve for:

severe symptomatic anemia

hemodynamic instability

critical Hb

Iron replacement still required afterward.

33. High-Yield Modern Pearls

Pearl 1

More oral iron does NOT necessarily mean more absorption.

Pearl 2

Alternate-day dosing often outperforms TID dosing.

Pearl 3

Ferritin may be falsely elevated in inflammation.

Pearl 4

Failure to investigate source of iron deficiency is dangerous.

Pearl 5

Most dialysis patients benefit more from IV iron than oral iron.

34. Practical Clinical Algorithms

Standard Adult Outpatient

Start:

Ferrous sulfate:

65 mg elemental iron

daily or alternate day

Recheck:

Hb in: 2–4 weeks.

If GI intolerance

reduce dose

alternate-day dosing

switch formulation

If poor response

Evaluate:

adherence

bleeding

malabsorption

inflammation

CKD

wrong diagnosis

CKD/Dialysis Patient

Assess:

Hb

ferritin

TSAT

Hemodialysis:

Usually: IV iron $\pm$ ESA.

Non-dialysis CKD:

oral or IV depending on response and severity.

35. Common Mistakes

Mistake 1

Excessive divided oral dosing.

Mistake 2

Stopping therapy too early.

Mistake 3

Ignoring GI malignancy risk.

Mistake 4

Using serum iron alone diagnostically.

Mistake 5

Giving ESA without adequate iron stores in CKD.

36. Evidence-Based Bottom Line

Best Oral Iron Strategy for Most Adults

Ferrous sulfate: 40–80 mg elemental iron:

once daily OR

alternate day.

Continue Treatment

For: ~3 months after Hb normalization.

Switch to IV Iron When

oral intolerance

malabsorption

active inflammation

CKD/dialysis

urgent repletion required.

37. Very Short Exam-Oriented Summary

First-line oral therapy

Ferrous sulfate.

Modern preferred dosing

Low-dose daily or alternate-day dosing.

Best diagnostic test

Ferritin.

Most important CKD principle

Functional iron deficiency due to hepcidin excess.

Preferred dialysis strategy

IV iron $\pm$ ESA.

Common CKD thresholds

Treat when:

TSAT $\leq 30\%$

ferritin ≤ 500 ng/mL.